Learned GNN Physics Simulator for Particle Jet Tagging

CS6012 Code-Based Project Presentation

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Based on J. Shlomi, P. Battaglia, J.-R. Vlimant, *Graph Neural Networks in Particle Physics*, arXiv:2007.13681

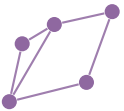
1. The Physics Problem: Why Graph Neural Networks?

What is Jet Tagging?

- Collisions produce quarks/gluons that hadronize into sprays of particles called **jets**.
- **Tagging**: Identifying the initiating particle from constituent kinematics.

Why Graphs over Images and Sequences?

- **CNNs (Images)**: Force continuous space onto pixel grids; $> 95\%$ cells are empty.
- **RNNs (Sequences)**: Impose arbitrary ordering (p_T sorting) on an unordered set.
- **GNNs (Point Clouds)**: Inherently respect **permutation invariance** and physical symmetries ($\Delta\eta, \Delta\phi, \Delta R$).



Particles = Nodes, Substructure = Edges

2. Model Architectures: No-Edge Set vs. Dynamic Graph

Model 1: Deep Sets Baseline

Philosophy: No edges ($E = \emptyset$).

$$y = \rho\left(\bigoplus_{i \in \mathcal{V}} \phi(x_i)\right)$$

- Per-particle MLP $\phi : \mathbb{R}^7 \to \mathbb{R}^{128}$
- Permutation-invariant masked pooling (Mean + Max)
- Classifier MLP $\rho(z)$
- **Complexity**: $O(N)$ operations

Model 2: Dynamic EdgeConv GNN

Philosophy: Relational message passing.

$$e_{ij} = \Theta([x_j - x_i,\, x_i]), \quad x_i' = \max_{j \in \mathcal{N}_k(i)} e_{ij}$$

- Dynamic k-NN in latent space ($k = 16$)
- EdgeConv blocks with residual shortcuts
- Vectorized in pure PyTorch (no C++ build fragility)
- **Complexity**: $O(k \cdot N)$ operations

Both models mathematically proven permutation-invariant to machine precision ($< 10^{-8}$).

3. Comparative Benchmark Results (100,000 Pythia 8.2 Jets)

Architecture	Acc	AUC	Latency	Param
Deep Sets	77.45%	0.8814	**0.03 ms**	**67k**
EdgeConv GNN	**80.65%**	**0.8830**	0.21 ms	108k

Key Insights:

- **Discriminative Power**: Dynamic EdgeConv captures relational substructure, reaching **80.65% accuracy** (+3.2% over Deep Sets) on 15,000 held-out Pythia test jets.
- **Efficiency Trade-off**: Deep Sets achieves **77.45% accuracy** while running 7× **faster** in inference (0.03 ms/jet) with 38% fewer parameters.

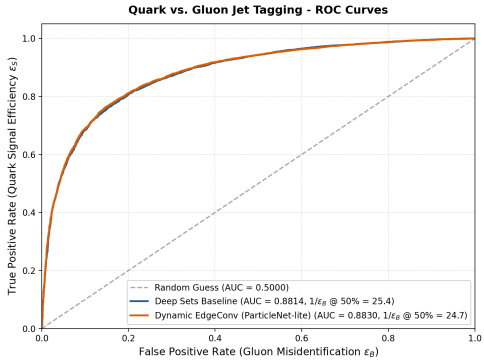


Figure: True Pythia 8.2 ROC Curves (100k Jets)

4. Canned Demo & Course Deliverables

Automated TA Canned Demo

- Single command: `./run_demo.sh`
- Runs out-of-the-box on CPU in **under 10 seconds**.
- Evaluates pre-trained checkpoints on bundled benchmark dataset.
- Outputs formatted comparison table and saves publication plot.

Submission Checklist

- Commented PyTorch codebase in GitHub repository.
- `README.txt` with signed Honor Code & LLM links.
- Pre-trained weights in `checkpoints/`.
- Final Project Report (PDF).
- 2–3 minute presentation video & slide deck.

Thank You!

GitHub Repository: `github.com/KraaK3N/gnn_jet_tagging`